

Tables

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Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_TAB_001	55	—	<pre> \begin{tabular}[t]{ l l } \hline \multicolumn{2}{ l }{Gravitational gauge field variables} \\ \hline \begin{tabular}[t]{ l l } \hline (h_{\mu}) & translational gauge field (h-field) \\ \hline (b^a_{\mu}) & determinant of h_a{}^{\mu} \\ \hline (A^{ab}_{\mu}) & inverse h-field \\ \hline (B_{\mu}) & rotational gauge field (A-field) \\ \hline (V_{\mu}) & WGT dilational gauge field (B-field) \\ \hline A^{\dagger ab}_{\mu} \equiv A^{ab}_{\mu} + 2\eta^{c[ab]}_{\mu} h_c{}^{\nu} V_{\nu} & eWGT dilational gauge field (V-field) \\ \hline \end{tabular} & eWGT extended rotational gauge field (A^{\dagger}-field) \\ \hline \end{tabular} Derivative operators D_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{ab}_{\mu} \Sigma_{ab}) \partial^a_{\mu} \equiv \partial_{\mu} + w B_{\mu} D^a_{\mu} \equiv h_a{}^{\mu} D^a_{\mu} \equiv h_a{}^{\mu} (\partial^a_{\mu} + \tfrac{1}{2} A^{ab}_{\mu} \Sigma_{ab}) \partial^{\dagger}_{\mu} \equiv \partial_{\mu} - w (V_{\mu} + \tfrac{1}{3} T_{\nu}) D^{\dagger}_{\mu} \equiv h_a{}^{\mu} D^{\dagger}_{\mu} \equiv h_a{}^{\mu} (\partial^{\dagger}_{\mu} + \tfrac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab}) D^2_{\mu} \equiv h_a{}^{\mu} D^2_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab}) Gauge field strengths \mathcal{R}^{ab}_{cd} = \mathcal{R}^{*ab}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}_{\nu]} + A^a{}_{c[\mu} A^{cb}_{\nu]}) \mathcal{R}^{\dagger ab}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}_{\nu]}) T^a{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]} T^*{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^a_{[\mu} b^a{}_{\nu]} T^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^{\dagger}_{[\mu} b^a{}_{\nu]} T^2{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^2_{[\mu} b^a{}_{\nu]} \mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]} \mathcal{H}^{\dagger}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \tfrac{1}{3} T_{\nu]}) Reduced A-fields and related quantities c^a{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]} c^*{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^a_{[\mu} b^a{}_{\nu]} c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^{\dagger}_{[\mu} b^a{}_{\nu]} {}^0A^{ab}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab}) {}^0A^{*ab}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c{}^{*ab}) {}^0A^{\dagger ab}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c{}^{\dagger ab}) Currents and related quantities \mathcal{L} \equiv h^{-1} L t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu} s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}_{\mu} j^a \equiv j^a{} b^{\mu}{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu} t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a V^c - s^{ac}{}_c V_b) j^{\dagger a} \equiv j^a - 2s^{ab}{}_b \tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu} \sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}_{\mu} \zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu} \tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a V^c - \sigma^{ac}{}_c V_b) \zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b </pre>		

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